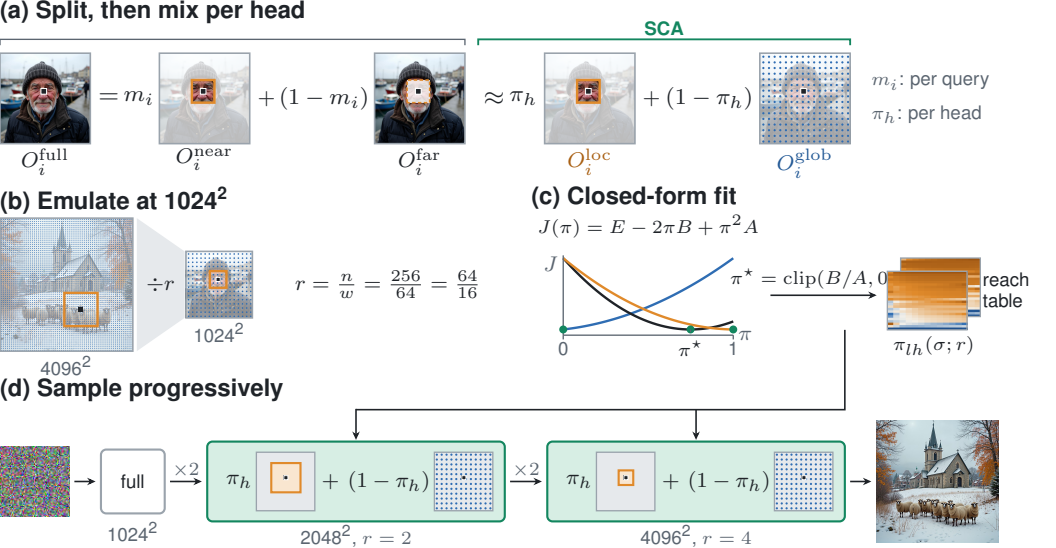

(a) Split, then mix per head
SCA
$= m_i$
$+ (1 - m_i)$
$\approx \pi_h$
$+ (1 - \pi_h)$
O_i^{full}
O_i^{near}
O_i^{far}
O_i^{loc}
O_i^{glob}
m_i: per query
π_h: per head
(b) Emulate at 1024^2
$\div r$
1024^2
4096^2
$r = \frac{n}{w} = \frac{256}{64} = \frac{64}{16}$
(c) Closed-form fit
$J(\pi) = E - 2\pi B + \pi^2 A$
J
$\pi^\star = \text{clip}(B/A, 0$
0
$\pi^\star$
1
π
reach
table
$\pi_{lh}(\sigma; r)$
(d) Sample progressively
full
$\times 2$
π_h
$+ (1 - \pi_h)$
$\times 2$
π_h
$+ (1 - \pi_h)$
1024^2
2048^2, $r = 2$
4096^2, $r = 4$